



BM1 Battery Monitor

Granz Scientific LLC

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1 Introduction

The BM1 Battery Monitor from Granz Scientific LLC is intended to provide a simple WiFi-connected solution to monitor vehicle and equipment batteries (or any other 12V or 24V systems). The device integrates with Home Assistant running on the local network and provides a convenient way to monitor batteries for over-discharge to protect them from damage. An on-board beeper is also configured to chirp if the battery voltage drops below a safe level for 12V or 24V lead-acid batteries, giving an audible warning. In addition to its primary function, the device provides a simple temperature sensor that is convenient for monitoring the environment.

NOTE: The device is not intended for direct exposure to water, and should be mounted in a dry location.

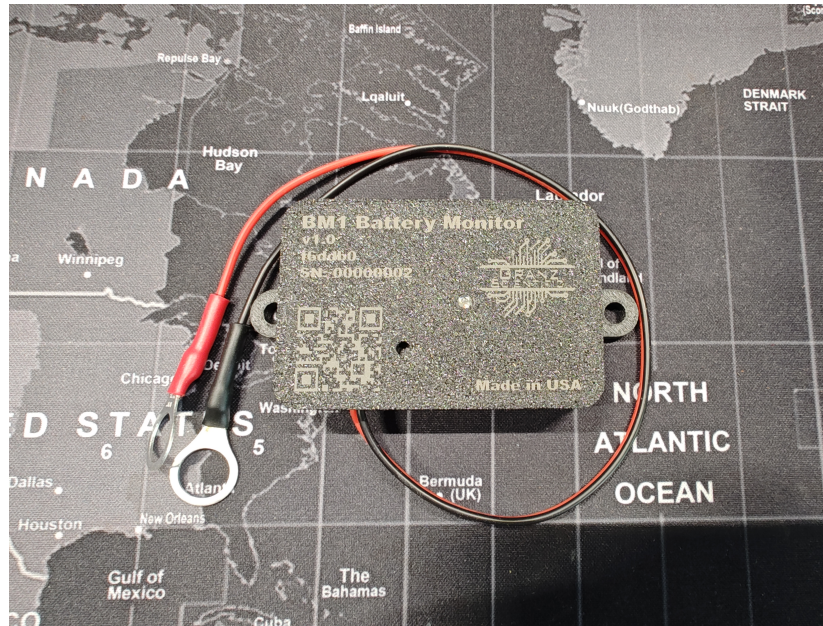


Figure 1: BM1 Battery Monitor

Upon integration into Home Assistant, the BM1 Battery Monitor will show as in Figure 2. Voltage and temperature graphs will show as in Figures 3 and 4, respectively.

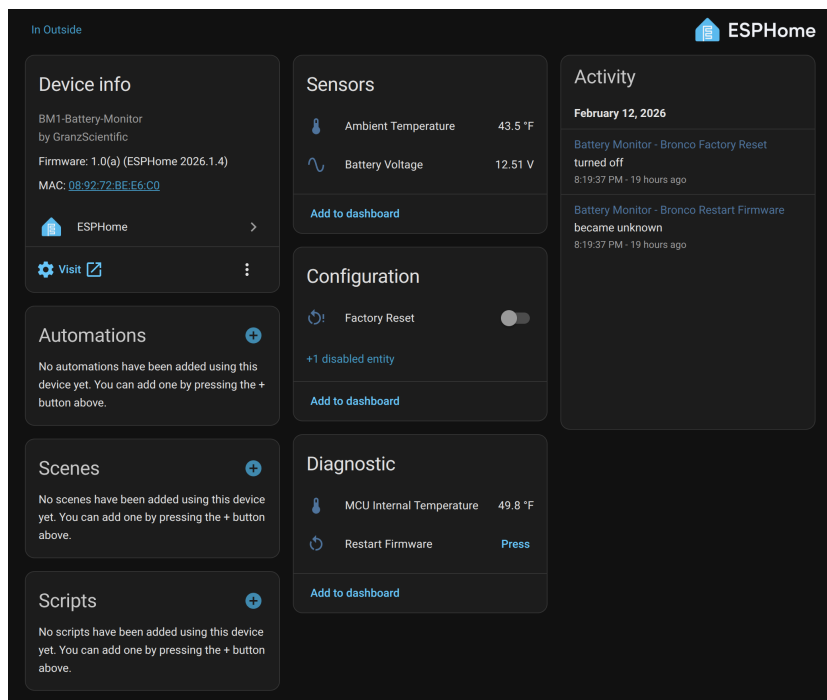


Figure 2: Home Assistant Device Page

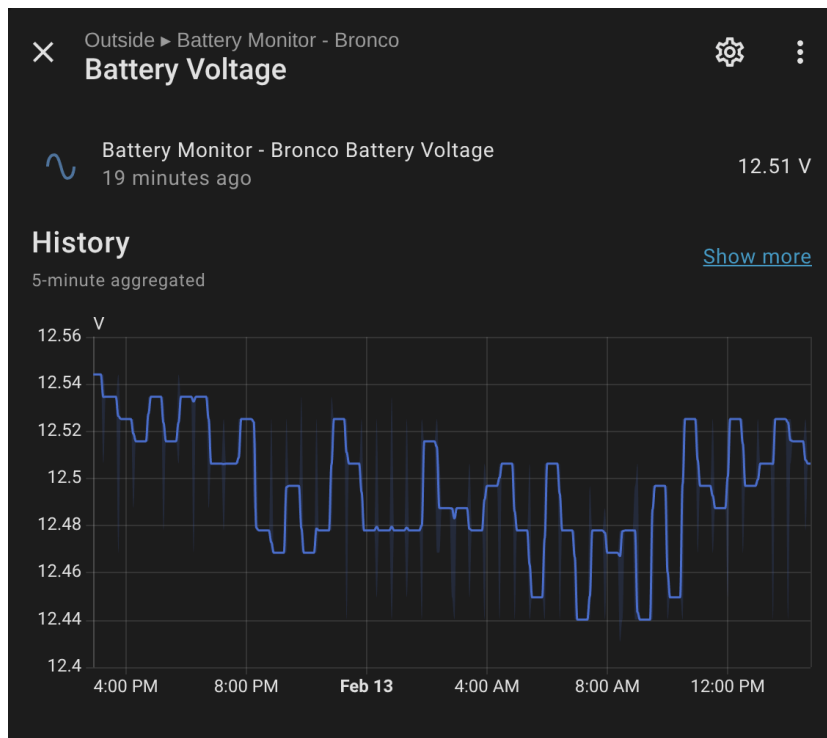


Figure 3: Voltage Plot Example

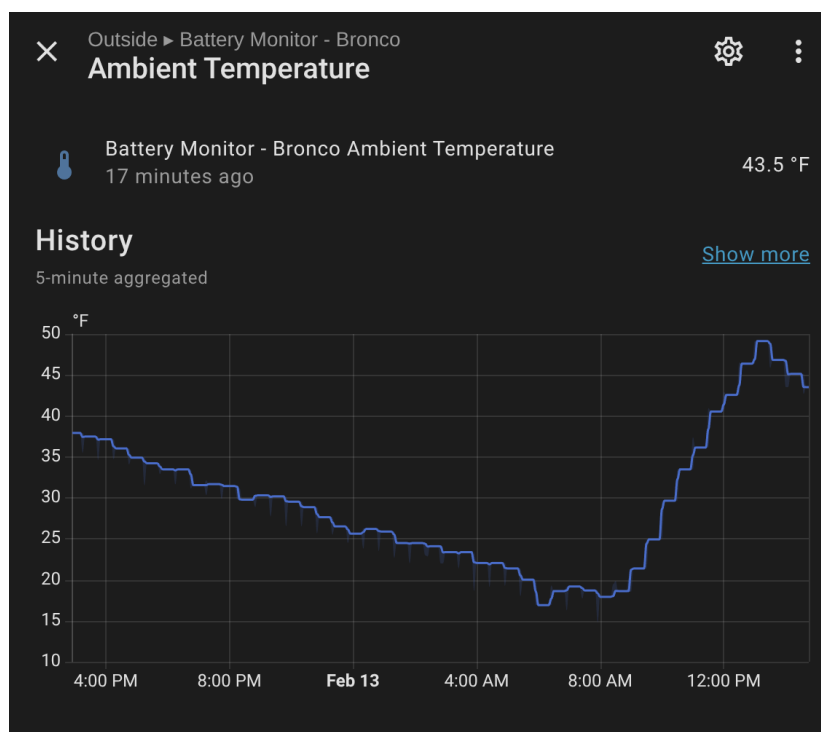


Figure 4: Temperature Plot Example

2 Specifications

Specification	Value	Units
Measurement Range	6 - 28	volts
Operating Temperature Range	-20 - +85	°C
Sample Interval	30	minutes
Sleep Current	~300	uAmperes
Active Time	~45	seconds
Active Current	~4	mAmperes
Average Current	~400	uAmperes
WiFi Working Range	150*	meters
Enclosure Size	64.5 x 42.5 x 18.5	millimeters
Mounting Tab Width	86.75	millimeters
Mounting Tab Hole Spacing	75.5	millimeters
Mounting Tab Hole Diameter	5	millimeters
Enclosure Material	Glass Fiber ABS	N/A

*Dependent on installed location, environment, and WiFi router

Table 1: Device specifications.

3 Device Setup

WARNING: After power is applied (connected to equipment battery) you will have 5 minutes to configure the device before it enters sleep mode.

Connect the device to the vehicle's battery, noting that the positive (red) wire goes to the positive terminal of the battery and the negative (black) wire goes to the negative terminal of the battery. The red indicator on the BM1 should light up and stay on. The device is now ready for setup.

Using a WiFi enabled device (phone, tablet, or computer) connect to the device's access point, which will show up with a name such as **"GS BM1 Battery Monitor f6ddb0"**. The last 6 characters of the name represent the last 6 hexadecimal digits of the MAC and will vary between individual devices. The default device WiFi access point password is **"12345678"**.

After connecting to the device's WiFi access point, opening a web browser should automatically take you to a setup page where you can enter your local network WiFi credentials. After clicking Save, the device will reboot and attempt to connect to your local network WiFi.

You will now have 5 minutes to complete the setup with the Home Assistant connection before the device goes to sleep to prevent draining the battery.

In Home Assistant, go to Settings → Devices & services and look for a pop-up asking if you want to add the new device. After clicking to add the device, it will be added to your Home Assistant ESPHome devices and ready for use. Keep in mind that the device will take a few samples at this point, and then immediately go to sleep for 30 minutes.

If you have any issues, or your local WiFi has changed, you may want to factory reset the device to allow initial setup again. To perform a factory reset and clear any saved settings, including WiFi credentials, the power connection must be cycled at least 5 times. Do this by disconnecting one power wire, then touch it to the battery terminal for a few seconds, remove it for a few seconds, and repeat five times. The device is back in configuration mode when the red indicator stays on solid for up to 5 minutes.

4 Customization

By default, the device does not use encryption for the API connection or have an OTA update password. **It is recommended that all users import the device into their environment (ESPHome-Builder or ESPHome CLI) and update the device to use encryption for security purposes.**

First, you will need to add a flag to Home Assistant which the device will pickup and which will prevent it from going to sleep every 30 minutes. This is important to be able to update the firmware wirelessly, since when the device is sleeping between samples it will be offline.

WARNING: When the flag is set to “on” in Home Assistant, the device (and other GS devices) will not go to sleep, potentially draining the battery faster than normal while left in this state. After you are finished with your updates, be sure to set the flag back to “off”.

Add the following to your Home Assistant configuration.yaml:

```
input_boolean:
  ota_update_mode:
    name: "OTA Update Mode"
    initial: off
    icon: mdi:chip
```



Figure 5: OTA Update Mode Switch

You will need to restart Home Assistant afterwards so that it will reload the configuration.yaml file. **After Home Assistant has restarted, add the entity “OTA Update Mode” to a dashboard and turn the switch to “on”.** It will show up as a switch as shown in Figure 5. Please note that if the device is currently sleeping, it may take up to 30 minutes before the device is online and stays awake, depending on when the last sample was taken and where it is in its sleep cycle.

There are two options for customizing the ESPHome firmware:

- **Install the ESPHome Device Builder add-on in Home Assistant** (recommended)
The device should automatically show up in ESPHomeBuilder and give you the option to “Take Control”. This will generate a YAML file for you which references the github repository.
- **Install ESPHome CLI**
You’ll need to clone the project git repository and then compile and upload the firmware manually.